

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260519_225244_b
- started_at_local: 2026-05-19 22:52:44
- finished_at_local: 2026-05-19 23:24:55
- duration_hhmm: 00:32
- duration_seconds: 1931.523
- seed_offset: 1000
- replicate_label: b
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.868026	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.90359	0.445	False	0.010863	0.719721
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.046321	0.115856	0.84925	0.445	False	0.00897	0.423769
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.049737	0.118013	0.870615	0.445	False	0.00837	0.677344

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.0	0.44	False	0.004297	1.374482
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.812304	0.44	False	0.003448	0.276204

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.868026, complexity 0.56, and adaptation efficiency -0.028692.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.90359, complexity 0.445, and adaptation efficiency 0.03612.
- Last-epoch runtime 54.69495 ms and distance evaluations 4648.0.
- Validation: surviving False, transplant delta 0.010863, positive scaffolds 1, Pareto runtime inflation 0.719721.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.844848, "gap_delta": -0.002615, "runtime_inflation": 0.719721, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.84925, complexity 0.445, and adaptation efficiency 0.008613.
- Last-epoch runtime 54.953275 ms and distance evaluations 4451.25.

- Validation: surviving False, transplant delta 0.00897, positive scaffolds 1, Pareto runtime inflation 0.423769.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.795701, "gap_delta": -0.005462, "runtime_inflation": 0.423769, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.049737, combined 0.118013.
- Accepted novelty 0.870615, complexity 0.445, and adaptation efficiency 0.006611.
- Last-epoch runtime 55.457125 ms and distance evaluations 4816.25.
- Validation: surviving False, transplant delta 0.00837, positive scaffolds 1, Pareto runtime inflation 0.677344.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.804735, "gap_delta": -0.005462, "runtime_inflation": 0.677344, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.0, complexity 0.44, and adaptation efficiency 0.0.
- Last-epoch runtime 66.69595 ms and distance evaluations 4451.25.
- Validation: surviving False, transplant delta 0.004297, positive scaffolds 2, Pareto runtime inflation 1.374482.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 1.609585, "gap_delta": -0.003363, "runtime_inflation": 1.374482, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.812304, complexity 0.44, and adaptation efficiency 0.014374.
- Last-epoch runtime 82.2059 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003448, positive scaffolds 1, Pareto runtime inflation 0.276204.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.216542, "gap_delta": 0.000224, "runtime_inflation": 0.276204, "same_gap_faster": false}

Judge Appendix

Conservative Interpretation of TSP Modular-Operator Discovery Results

Main Transfer Evidence (Held-out TSPLIB & Family Holdout Gaps)

- ****Held-out TSPLIB Gap:**** No improvement over baseline (all have gap delta = 0.0).
- ****Family Holdout Gap:****

- The modular operator conditions related to restart controllers (phase7_modular_operator_*) show modest family gap improvements (~ -0.004 to -0.007).
- Families with positive signal: *clustered_tsp* and *two_cluster_bottleneck_tsp*.
- No severe regressions observed in any operators.

Validation via Transplant and Ablation Checks

- All modular restart controller operators fail clean transplantation into some scaffolds (sparse_three_opt, clustered_local_search).
- Transplant mean gap deltas for restart controllers are slightly positive (performance worsens) overall ($\sim +0.008$ to $+0.011$), indicating limited transplant robustness.
- Ablation signals for restart controllers show small gap increases when disabled ($\sim +0.0026$ to $+0.0054$), indicating some operator effect but weak.
- The candidate pruner operator:
 - Shows no improvement on held-out families or held-out TSPLIB.
 - Ablation indicates disabling does not degrade performance (gap delta slightly negative).
 - Does not transplant cleanly into clustered_local_search.

Operator Complexity and Runtime Tradeoffs (Pareto)

- Restart controller operators have moderate complexity (~ 12 lines, score 0.44).
- Runtime inflation for restart controllers is high (0.42 to 1.37), with no better gap at same runtime.
- Candidate pruner exhibits low complexity with negligible gap improvement but also runtime inflation (~ 0.28).
- No operator shows Pareto improvements (better gap at same or less runtime).

Claims on Reusable Operator Structure

- Restart controllers modify restart pressure using deterministic triggers under stagnation — a compact reusable mechanism.
- Candidate pruner adapts candidate list size based on instance descriptors, offering a potentially reusable neighborhood control.
- However, limited transplant success and weak ablation signals reduce confidence in robust reusable operator structure.

Summary and Conclusion

Operator (Condition Name)	Transfer Gap Delta	Family Gap Delta	Ablation Support	Transplant Success	Pareto Improvement	Surviving Candidate
phase7_modular_operator_evolution	-0.0055 approx.	-0.0074 approx.	Weak (small gap increase if disabled)	Partial (fails some scaffolds)	None (runtime inflation)	No
phase7_modular_operator_random_replay	-0.0055 approx.	-0.0074 approx.	Weak	Partial	None	No

Operator (Condition Name)	Transfer Gap Delta	Family Gap Delta	Ablation Support	Transplant Success	Pareto Improvement	Surviving Candidate
phase7_modular_operator_diversity_residual_replay	-0.0045 approx.	-0.0045 approx.	Weak	Partial	None	No
phase7_modular_operator_compression_pressure	-0.0035 approx.	-0.0035 approx.	Weak	Partial	None (high runtime inflation)	No
phase7_modular_operator_pareto_selection	+0.00009	Slight regression	No ablation signal	Partial	None	No

- None of the modular operators survive all conditions (transplant, ablation, family transfer, held-out TSPLIB).
- The restart controller class shows some promise as a reusable component for clustered and bottleneck TSP families but fails transplant and Pareto criteria.
- No new algorithm discovery supported by the modular-operator approach due to failure in scaffold and family holdout tests.
- Full solver evolution (phase7_full_solver_evolution) achieves accepted epochs but no modular validated reusable operator extracted.

Final Conservative Conclusion

- ****No modular operator qualifies as a surviving reusable operator given transplant and ablation validation.****
- Restart controllers exhibit modest family transfer gains and weak ablation support but fail clean transplant and Pareto tradeoffs.
- Candidate pruner does not show meaningful transfer or ablation signal.
- Held-out TSPLIB gaps remain equal or worse than baseline for all operators, indicating no generalization to new problem sets.
- No modular operator discovery claim or reusable operator extraction can be made currently.
- The best evidence points to limited reusable restart pressure control potential within clustered TSP families, but this requires further development and stronger validation for practical use.